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RESEARCH

Inverse scattering transform and soliton solutions for the coupled Hirota-Maxwell-Bloch system with nonzero background

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Abstract

In this paper, we study the N-soliton solutions for Hirota-Maxwell-Bloch system (H-MB system) via Riemann-Hilbert approach. By a Lax pair of this system, the inverse scattering method is employed to construct a Riemann-Hilbert problem (RHP) for the H-MB system under a non-zero background, which is imposed only on the η component (with u, $p \to 0$ as $x \to \pm\infty$). From this RHP, the long-time asymptotics of the soliton is derived, and the N-soliton solutions are obtained, with explicit expressions given for the cases $N = 1, 2, 3$. Furthermore, the dynamical properties of the case where $N = 1, 2$ are investigated, and the corresponding three-dimensional plots are presented. Compared with the conclusions reported under specific parameter choices, the results obtained in this paper are valid for general parameters α and β. This is of great significance for the study of nonlinear pulse propagation in optical fibers.

Keywords Hirota-Maxwell-Bloch system · Solitons · Lax pair · Inverse scattering method · Riemann-Hilbert problem

1 Introduction

The investigation about nonlinear phenomena occupies a central position within contemporary theoretical and mathematical physics. A substantial number of fundamental physical laws are expressed through nonlinear partial differential equations (NLPDEs), which govern systems across diverse domains, including fluid dynamics, plasma physics, nonlinear optics, condensed matter physics, and even biophysics [1, 2]. In contrast to their linear counterparts, NLPDEs frequently give rise to intricate and complex behaviors, notably the emergence of stable, localized wave structures capable of propagating without altering their form or speed. Among the most notable of such structures are solitons, which originate from a precise balance between nonlinear effects and dispersion. Initially observed by John Scott Russell in 1834 as a "wave of translation" in a shallow water channel, the soliton concept was subsequently placed on a rigorous mathematical footing with the formulation of the Korteweg-de Vries (KdV) equation.

The inverse scattering transform (IST) emergenced in the late 1960s marked a turning point in the study of NLPDEs, offering a systematic framework for solving initial value problems associated with a family of equations now referred to as integrable systems [3]. The IST can be interpreted as a nonlinear counterpart to the Fourier transform employed in linear equations: it maps the evolution of initial data onto a simplified linear evolution of associated scattering data, from which the solution may be reconstructed at arbitrary later times. This advancement paved the way for the identification of a broad class of integrable equations alongside their exact solutions, most notably the N-soliton solutions that capture the interactions among multiple solitons [4, 5]. In the ensuing decades, a range of alternative powerful methods has emerged for constructing exact solutions to integrable NLPDEs. These include the Hirota bilinear method, which reformulates the NLPDE into a more tractable bilinear form [6, 7], Bäcklund transformations, which enable the generation of new solutions from pre-existing ones [8], and Darboux transformations, which offer an algebraic means of solution construction. Such approaches have proved instrumental in uncovering a diverse array of exact solutions encompassing multi-soliton solutions [9], breather solutions [10], and lump solutions [7], thereby deepening our comprehension of nonlinear wave phenomena.

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Among these analytical methodologies, the Riemann-Hilbert (RH) approach has come to be recognized as an especially elegant and rigorous framework for addressing integrable systems. The Riemann-Hilbert problem (RHP) is to find a matrix function, it is analytic on the complex plane, and satisfies a prescribed jump condition along a given contour. The link between integrable systems and the RHP was forged via a reformulation of the inverse scattering transform [11]. From this examination, we can get the jump matrix and scattering data. Through the direct scattering process, the analytic properties of the eigenfunctions of the Lax pair are understood. The IST problem is transformed into a matrix RHP. Solving this problem then allows for the recovery of the potential function, namely, the solution to the NLPDE [12].

The RH methodology presents several notable benefits relative to the classical IST approach, which is founded upon Gel'fand-Levitan-Marchenko integral equations. It affords a more direct and unifying framework capable of accommodating a broader class of problems, including scenarios involving non-zero boundary conditions [13, 14], longtime asymptotic analysis [15], and solutions associated with higher order poles in the scattering data [1, 16]. Specifically, this methodology has been successfully applied to investigate the Cauchy problems with step-like initial data, pole solutions, and the dynamic behaviors of soliton solutions for various integrable systems, including the nonlocal Hirota equation, the nonlocal combined nonlinear Schrödinger-Gerdjikov-Ivanov model, and three-coupled Lakshmanan-Porsezian-Daniel model [17–19]. Moreover, its rigorous mathematical underpinnings have established it as an indispensable tool not only within the realm of integrable systems but also across ostensibly disparate fields such as random matrix theory [11, 20, 21]. Within this framework, the derivation of N-soliton solutions is rendered particularly efficient. In the reflectionless scenario, wherein the jump matrix becomes trivial, the problem of solving the RHP is transformed into that of solving a system of linear equations, from which the N-soliton solution can be obtained, which typically appear in the form of determinants. [22].

The efficacy and adaptability of the RH framework have been evidenced through its successful implementation across a wide spectrum of integrable equations. This encompasses numerous members of the nonlinear Schrödinger (NLS) hierarchy, which serve as fundamental models in nonlinear optics, plasma physics, and water waves. For example, this method has been widely applied in deriving the N-soliton solutions of the generalized NLS equation [23], the Kundu-Eckhaus equation [15], the Kundu-type equation [24], the novel Kundu-NLS equation [25], the Sasa-Satsuma equation and its generalizations [26–28], the Fokas-Lenells equation [29], the Chen-Lee-Liu equation [30, 31], and higher-order NLS equations [16, 32]. The method has also been extended to multi-component systems [33–35], nonlocal equations [36], and equations from the KdV hierarchy, such as the matrix modified KdV equation [37] and coupled extended modified KdV equations [38]. Additional notable models addressed successfully through the RH methodology comprise the massive Thirring model [22], the Degasperis-Procesi equation [39], and the short pulse equation [40]. This substantial body of work attests to the robustness and far-reaching applicability of the RH method within the domain of integrable systems.

This paper primarily investigates the following Hirota-Maxwell-Bloch system (H-MB) system, which constitute an integrable system of significant physical importance:

$$\begin{aligned} u_t &= \mathrm{i}\alpha\left(\frac{1}{2}u_{xx} + |u|^2 u\right) + \beta\left(u_{xxx} + 6|u|^2 u_x\right) + 2p, \\ p_x &= 2\mathrm{i}\omega p + 2u\eta, \\ \eta_x &= -\left(up^* + u^* p\right), \end{aligned} \tag{1}$$

where $u = u(x,t)$, $p = p(x,t)$ and $\eta = \eta(x,t)$ are complex functions with $(x,t) \in \mathbb{R} \times \mathbb{R}_+$, α, β and ω are real constants, and the superscript '*' indicating complex conjugation. And u represents the electric field envelope, p is the polarization component, and η denotes the population inversion component. The physical significance of this system lies in its description of ultrashort optical pulse propagation within an erbium-doped nonlinear fiber. This model is a coupling of the Hirota and the Maxwell-Bloch equations. The H-MB system is of great interest in nonlinear optics and optical communications, as it captures essential physical effects that become important for femtosecond pulse propagation. Understanding the soliton solutions of this system is crucial for predicting and controlling the behavior of these ultrashort pulses, with potential applications in high-speed optical switching and data transmission.

While the Maxwell-Bloch equations have been studied using the RH approach [34], and related spin systems have also been investigated [41]. Studying the N-soliton solutions about the H-MB system using the RHP method is of great significance. The coupling between the field envelope (governed by the Hirota equation) and the atomic variables (governed by the Maxwell-Bloch equations) introduces additional complexity into the spectral problem and the subsequent analysis. The primary motivation of this work is to apply the rigorous and systematic RH approach to the H-MB system to derive its N-soliton solutions explicitly. This endeavor not only extends the applicability of the RH method to another important physical model but also provides a deeper understanding of the nonlinear wave dynamics within this coupled system. We study the H-MB system under the following conditions:

$$u = 0,\ p = 0,\ \eta = 1,\ x \to \pm\infty. \tag{2}$$

Compared with existing studies on the H-MB system, which predominantly focus on soliton solutions under a zero background [42], this work lies in the successful construction of N-soliton solutions under a nonzero background specifically imposed on the η component using the RH approach. This provides a more realistic representation of ultrashort pulse propagation in active optical fibers. While Darboux transformations are commonly used to find soliton solutions, the RH approach employed here provides a more comprehensive analytical framework. It not only allows for the derivation of explicit N-soliton solutions but also facilitates the rigorous analysis of scattering data symmetries and, crucially, the long-time asymptotic behavior, which is often absent in studies utilizing other methods.

The structure of this paper is as follows: Sects. 2.1 and 2.2, we presents the Lax pair, Jost solutions, the scattering matrix, and the adjoint Jost solutions. Section 2.3 constructs the RHP and considers the solutions of the H-MB system by regular and non-regular. Section 3 calculates the asymptotic behavior of the solutions to the H-MB system. Section 4 provides the explicit expressions for the N-soliton solutions, along with corresponding graphical illustrations of the soliton solutions. Section 5 computes the infinite conservation laws associated with the H-MB system. Finally, Sect. 6 presents the conclusion of this article.

2 Inverse scattering transform method

2.1 Hirota–Maxwell–Bloch system

As an integrable system within the multi-component AKNS framework [43], Eq. (1) admit a Lax pair of the following form:

$$\begin{cases} Y_x = UY, \quad U = -\mathrm{i}\lambda\sigma_3 + U_0, \\ Y_t = VY, \quad V = \lambda^3 V_3 + \lambda^2 V_2 + \lambda V_1 + V_0 + \mathrm{i}\dfrac{1}{\lambda+\omega} V_{-1}, \end{cases} \tag{3}$$

where $\lambda \in \mathbb{C}$ is the spectral parameter, $Y = Y(x,t,\lambda)$ is a matrix-valued function, $\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is the third Pauli matrix, and the potential matrices U_0, V_{-1}, V_0, V_1, V_2, V_3 are given by

$$U_0 = \begin{pmatrix} 0 & u \\ -u^* & 0 \end{pmatrix}, \quad V_{-1} = \begin{pmatrix} \eta & -p \\ -p^* & -\eta \end{pmatrix},$$

$$V_0 = \begin{pmatrix} \dfrac{\alpha}{2}\mathrm{i}|u|^2 - \beta(uu_t^* - u_t u^*) & 2\beta|u|^2 u + \dfrac{\alpha}{2}\mathrm{i}u_t + \beta u_{tt} \\ -2\beta|u|^2 u + \dfrac{\alpha}{2}\mathrm{i}u_t^* - \beta u_{tt}^* & -\dfrac{\alpha}{2}\mathrm{i}|u|^2 + \beta(uu_t^* - u_t u^*) \end{pmatrix},$$

$$V_1 = \begin{pmatrix} -2\beta\mathrm{i}|u|^2 & \alpha u - 2\beta\mathrm{i}u_t \\ -\alpha u^* - 2\beta\mathrm{i}u_t^* & 2\beta\mathrm{i}|u|^2 \end{pmatrix},$$

$$V_2 = \begin{pmatrix} -\alpha\mathrm{i} & -4\beta u \\ 4\beta u^* & \alpha\mathrm{i} \end{pmatrix}, \quad V_3 = \begin{pmatrix} 4\mathrm{i}\beta & 0 \\ 0 & -4\mathrm{i}\beta \end{pmatrix}.$$

That is to say, the compatibility condition $Y_{xt} = Y_{tx}$ (or $U_t - V_x + [U,V] = 0$, where $[U,V] = UV - VU$) of the Lax pair (3) holds if and only if the potential functions u, p, η satisfy Eq. (1). Notably, the matrix U_0 is anti-Hermitian and the trace is zero, i.e.

$$U_0^\dagger = -U_0, \quad \mathrm{tr}(U_0) = 0. \tag{4}$$

The asymptotic condition (2) implies that

$$Y(x,t,\lambda) = e^{-\theta(x,t,\lambda)\sigma_3}, \quad x \to \pm\infty, \tag{5}$$

where $\theta(x,t,\lambda) = \mathrm{i}\lambda x - \mathrm{i}\left(4\beta\lambda^3 - \alpha\lambda^2 + \frac{1}{\lambda+\omega}\right)t$. It is then natural to introduce the expression Y as

$$Y(x,t,\lambda) = J(x,t,\lambda)e^{-\theta(x,t,\lambda)\sigma_3}. \tag{6}$$

Substituting this transformation into (3) yields

$$J_x = -\mathrm{i}\lambda[\sigma_3, J] + U_0 J, \tag{7}$$

and

$$J_t = (4\beta\lambda^3\mathrm{i} - \alpha\lambda^2\mathrm{i} + \frac{\mathrm{i}}{\lambda+\omega})[\sigma_3, J] + Q_0 J, \tag{8}$$

where $Q_0 = \alpha\lambda^2\mathrm{i}\sigma_3 + \lambda^2 V_2 + \lambda V_1 + V_0 + \frac{\mathrm{i}}{\lambda+\omega}V_{-1} - \frac{\mathrm{i}}{\lambda+\omega}\sigma_3$.

2.2 Riemann–Hilbert formulation

To formulate the RHP for the H-MB system (1), the matrix-valued functions $J_\pm$ are introduced as two solutions to Eq. (7):

$$J_- = \left([J_-]_1, [J_-]_2\right), \tag{9}$$

$$J_+ = \left([J_+]_1, [J_+]_2\right). \tag{10}$$

They satisfy

$$J_\pm(x,\lambda) \to I, \quad x \to \pm\infty, \tag{11}$$

where let $[J_\pm]_1$ and $[J_\pm]_2$ be the first and second columns of $J\pm$, and I is the identity matrix. Introduce the matrix

$$E(x,\lambda) = e^{-\mathrm{i}\lambda\sigma_3 x}, \tag{12}$$

and the Jost functions

$$\Phi = J_- E, \quad \Psi = J_+ E. \tag{13}$$

It can be shown that Φ and Ψ are solutions of the spatial part of Lax pair (3), i.e.

$$Y_x + \mathrm{i}\lambda\sigma_3 Y = U_0 Y. \tag{14}$$

When Eq. (14) is a homogeneous, i.e. $U_0 Y = 0$, the solution is E. Combining this with the condition (2), we obtain the Volterra integral:

$$J_-(x,\lambda) = I + \int_{-\infty}^{x} e^{\mathrm{i}\lambda\sigma_3(y-x)} U_0(y) J_-(y,\lambda) e^{\mathrm{i}\lambda\sigma_3(x-y)} dy, \tag{15}$$

$$J_+(x,\lambda) = I - \int_{x}^{\infty} e^{\mathrm{i}\lambda\sigma_3(y-x)} U_0(y) J_+(y,\lambda) e^{\mathrm{i}\lambda\sigma_3(x-y)} dy. \tag{16}$$

For instance, setting $[J_-]1 = (\varphi_1, \varphi_2)^T$, we have

$$\varphi_1(x,\lambda) = 1 + \int_{-\infty}^{x} u(y)\varphi_2(y,\lambda) dy, \tag{17}$$

$$\varphi_2(x,\lambda) = -\int_{-\infty}^{x} u^*(y)\varphi_1(y,\lambda) e^{2\mathrm{i}\lambda(x-y)} dy. \tag{18}$$

Thus, the analyticity of $J_\pm$ are given

- $[J_-]_1$ and $[J_+]_2$ are analytic in $D^+ = \{\lambda \in \mathbb{C} \mid \mathrm{Im}\,\lambda > 0\}$, and can be continuously extended to real axis $\mathbb{R}$.
- $[J_+]_1$ and $[J_-]_2$ are analytic in $D^- = \{\lambda \in \mathbb{C} \mid \mathrm{Im}\,\lambda < 0\}$, and can be continuously extended to real axis $\mathbb{R}$.

Since both $\Phi(x,\lambda)$ and $\Psi(x,\lambda)$ are the solutions of Eq. (13). So, existing $S(\lambda)$ satisfies

$$\Phi(x,\lambda) = \Psi(x,\lambda) S(\lambda), \quad \lambda \in \mathbb{R}, \tag{19}$$

i.e.

$$J_- E = J_+ E S, \quad \lambda \in \mathbb{R}. \tag{20}$$

Writing $\Phi(x,\lambda)$ and $\Psi(x,\lambda)$ as column vectors

$$\Phi = (\phi_1, \phi_2), \quad \Psi = (\psi_1, \psi_2). \tag{21}$$

By Liouville's identity and combining with (4), we obtain

$$\det J_\pm(x,\lambda) = 1. \tag{22}$$

Then the relation (19) implies that

$$\det S(\lambda) = 1, \quad \lambda \in \mathbb{R}. \tag{23}$$

Define

$$P^+ = \left([J_-]_1, [J_+]_2\right) = J_- H_1 + J_+ H_2. \tag{24}$$

For $\lambda \in \mathbb{C}_+$, it is analytic, where

$$H_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad H_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \tag{25}$$

Also, (15)–(16) yields that for $\lambda \in \mathbb{C}_+$,

$$P^+ \to I, \quad \lambda \to \infty. \tag{26}$$

On the other hand, the analogue of P^- in the region $\mathbb{C}_-$ is constructed by setting

$$J_-^{-1} = \left(\begin{bmatrix} J_-^{-1} \end{bmatrix}_1, \begin{bmatrix} J_-^{-1} \end{bmatrix}_2\right), \quad J_+^{-1} = \left(\begin{bmatrix} J_+^{-1} \end{bmatrix}_1, \begin{bmatrix} J_+^{-1} \end{bmatrix}_2\right). \tag{27}$$

Then, using the relation

$$0 = \left(J J^{-1}\right)_x = J_x J^{-1} + J\left(J^{-1}\right)_x, \tag{28}$$

and finding matrix-valued functions $J_\pm^{-1}$ satisfy the adjoint scattering problem corresponding to (7):

$$K_x = -\mathrm{i}\lambda[\sigma_3, K] - K U_0. \tag{29}$$

Similarly, the analytic of $J_\pm^{-1}$ are as follows:

- $\left[J_+^{-1}\right]_1$ and $\left[J_-^{-1}\right]_2$ are analytic in $D^+ = \{\lambda \mid \mathrm{Im}\,\lambda > 0\}$, and can be continuously extended to real axis $\mathbb{R}$.
- $\left[J_-^{-1}\right]_1$ and $\left[J_+^{-1}\right]_2$ are analytic in $D^- = \{\lambda \mid \mathrm{Im}\,\lambda < 0\}$, and can be continuously extended to real axis $\mathbb{R}$.

Writing the inverses $\Phi^{-1}(x,\lambda)$ and $\Psi^{-1}(x,\lambda)$ as row vectors

$$\Phi^{-1} = \begin{pmatrix} \widehat{\phi}_1 \\ \widehat{\phi}_2 \end{pmatrix}, \quad \Psi^{-1} = \begin{pmatrix} \widehat{\psi}_1 \\ \widehat{\psi}_2 \end{pmatrix}. \tag{30}$$

Then, according to Eq. (20)

$$E^{-1} J_-^{-1} = R(\lambda) E^{-1} J_+^{-1}, \quad \lambda \in \mathbb{R}, \tag{31}$$

where $R(\lambda) = (r_{ij})_{2\times 2} = S^{-1}$. Define

$$P^- = \left(\begin{bmatrix} J_-^{-1} \end{bmatrix}_1, \begin{bmatrix} J_+^{-1} \end{bmatrix}_2\right) = H_1 J_-^{-1} + H_2 J_+^{-1}, \tag{32}$$

which is analytic for $\lambda \in \mathbb{C}-$ and satisfies the asymptotic condition as follows:

$$P^- \to I, \quad \lambda \in \mathbb{C}-, \ \lambda \to \infty. \tag{33}$$

In summary, the matrix-valued functions Φ, Ψ, Φ^{-1}, and Ψ^{-1} expressed according to their regions of analyticity as follows:

$$\Phi = (\phi_1^+, \phi_2^-), \quad \Psi = (\psi_1^-, \psi_2^+), \tag{34}$$

$$\Phi^{-1} = \begin{pmatrix} \widehat{\phi}_1^- \\ \widehat{\phi}_2^+ \end{pmatrix}, \quad \Psi^{-1} = \begin{pmatrix} \widehat{\psi}_1^+ \\ \widehat{\psi}_2^- \end{pmatrix}, \tag{35}$$

where the superscripts '$\pm$' indicate the respective half-planes of analyticity. And the Eq. (19) implies that

$$S = \Psi^{-1}\Phi = \begin{pmatrix} \widehat{\psi}_1^+ \\ \widehat{\psi}_2^- \end{pmatrix} (\phi_1^+, \phi_2^-), \tag{36}$$

$$R = \Phi^{-1}\Psi = \begin{pmatrix} \widehat{\phi}_1^- \\ \widehat{\phi}_2^+ \end{pmatrix} (\psi_1^-, \psi_2^+). \tag{37}$$

Hence, the analytic properties of S and R be readily deduced from Eqs. (36)–(37) as follows:

- s_{11} and s_{22} are analytic in $\mathbb{C}+$ and $\mathbb{C}-$, respectively.
- r_{11} and r_{22} are analytic in $\mathbb{C}-$ and $\mathbb{C}+$, respectively.
- s_{12}, s_{21}, r_{12} and r_{21} are analytic only on $\mathbb{R}$.

So the matrices S and R can be written as

$$S = \begin{pmatrix} s_{11}^+ & s_{12} \\ s_{21} & s_{22}^- \end{pmatrix}, \quad R = \begin{pmatrix} r_{11}^- & r_{12} \\ r_{21} & r_{22}^+ \end{pmatrix}. \tag{38}$$

Furthermore, (19) and (38) imply the following relationships between s_{ij} and r_{ij}:

$$r_{11} = s_{22}, \quad r_{22} = s_{11}, \quad r_{12} = -s_{12}, \quad r_{21} = -s_{21}. \tag{39}$$

In conclusion, if we define

$$P(x, \lambda) = P^{\pm}(x, \lambda), \quad \lambda \in \mathbb{C}_{\pm}, \tag{40}$$

then the RHP for $P(x, \lambda)$ is obtained as follows:

Lemma 1 *Exist a* 2×2 *complex matrix-valued function* $P(x, \lambda)$ *that satisfies:*

- *In* $\mathbb{C}\backslash\mathbb{R}$, $P(x, \lambda)$ *is analytic and possesses nontangential limits from both sides of* $\mathbb{R}$.
- *On* $\mathbb{R}$, *the boundary values satisfy*

$$P^-(x, \lambda)P^+(x, \lambda) = G(x, \lambda), \quad \lambda \in \mathbb{R}, \tag{41}$$

where

$$G = E(H_1 + H_2 S)(H_1 + S^{-1}H_2)E^{-1} = E \begin{pmatrix} 1 & r_{12} \\ s_{21} & 1 \end{pmatrix} E^{-1}. \tag{42}$$

- *Normalization condition: for* $\lambda \in \mathbb{C}\backslash\mathbb{R}$,

$$P(x, \lambda) \to I, \quad \lambda \to \infty. \tag{43}$$

In addition, from the normalization condition (43), we can assume that

$$P^{\pm}(x, \lambda) = I + \lambda^{-1} P_1^{\pm}(x) + O(\lambda^{-2}), \quad \lambda \to \infty, \quad \lambda \in \mathbb{C}\pm, \tag{44}$$

where $P_1^{\pm}(x)$ is independent of λ. Substituting (44) into (7) and (29) and then equating coefficients of like powers of λ yields

$$\lambda^{-1}: \quad P_{1,x}^+ = U_0 P_1^+, \tag{45}$$

$$\lambda: \quad 0 = -\mathrm{i}[\sigma_3, I], \tag{46}$$

$$\lambda^0: \quad 0 = -\mathrm{i}[\sigma_3, P_1^+] + U_0, \qquad 0 = -\mathrm{i}[\sigma_3, P_1^-] - U_0. \tag{47}$$

Hence, $P_1^+(x)$ be expressed as

$$P_1^+(x) = \frac{1}{2\mathrm{i}} \begin{pmatrix} \int_{-\infty}^x (|u(y)|)^2\, dy & u(x) \\ u^*(x) & \int_x^{\infty} (|u(y)|)^2\, dy \end{pmatrix}, \tag{48}$$

and the potential function u can be constructed as

$$u = 2\mathrm{i}(P_1^+)_{12} = -2\mathrm{i}(P_1^-)_{12}. \tag{49}$$

Thus, according to Eq. (1), the expressions for p and η can be obtained as follows:

$$\begin{aligned} p =& \mathrm{i}(P_1^+)_{12,t} + \frac{\alpha}{2}(P_1^+)_{12,xx} - \beta\mathrm{i}(P_1^+)_{12,xxx} \\ & - 24\beta\mathrm{i}(P_1^+)_{12,x}|(P_1^+)_{12}|^2 + 4\alpha(P_1^+)_{12}|(P_1^+)_{12}|^2, \end{aligned} \tag{50}$$

$$\begin{aligned} \eta =& \frac{1}{4\mathrm{i}(P_1^+)_{12}}(\mathrm{i}(P_1^+)_{12,t} + \frac{\alpha}{2}(P_1^+)_{12,xx} - \beta\mathrm{i}(P_1^+)_{12,xxx} \\ & - 24\beta\mathrm{i}(P_1^+)_{12,x}|(P_1^+)_{12}|^2 + 4\alpha(P_1^+)_{12}|(P_1^+)_{12}|^2)_x \\ & - \frac{\omega}{2(P_1^+)_{12}}(\mathrm{i}(P_1^+)_{12,t} + \frac{\alpha}{2}(P_1^+)_{12,xx} - \beta\mathrm{i}(P_1^+)_{12,xxx} \\ & - 24\beta\mathrm{i}(P_1^+)_{12,x}|(P_1^+)_{12}|^2) - 2\alpha\omega|(P_1^+)_{12}|^2. \end{aligned} \tag{51}$$

Furthermore, using relationship of (20), (24) and (32) yields

$$\det P^+ = r_{22} = s_{11}, \quad \det P^- = s_{22} = r_{11}. \tag{52}$$

Combining this with (43), as $\lambda \to \infty$, obtain

$$r_{22}(\lambda) \to 1 + \frac{1}{2\mathrm{i}\lambda}\int_{-\infty}^{\infty} |u(y)|^2 dy + O(\lambda^{-2}), \quad \lambda \in \mathbb{C}_+, \tag{53}$$

$$s_{22}(\lambda) \to 1 - \frac{1}{2\mathrm{i}\lambda}\int_{-\infty}^{\infty} |u(y)|^2 dy + O(\lambda^{-2}), \quad \lambda \in \mathbb{C}_-. \tag{54}$$

Moreover, combining (4), (7) and (29) gives

$$J_{\pm}^{\dagger}(\lambda^*)_x = -\mathrm{i}\lambda^*[\sigma_3, J_{\pm}^{\dagger}(\lambda^*)] - J_{\pm}^{\dagger}(\lambda^*)U_0, \tag{55}$$

$$J_{\pm}^{-1}(\lambda)_x = -\mathrm{i}\lambda[\sigma_3, J_{\pm}^{-1}(\lambda)] - J_{\pm}^{-1}(\lambda)U_0. \tag{56}$$

It is known that $J_{\pm}^{\dagger}(\lambda^*)$ and $J_{\pm}^{-1}(\lambda)$ satisfy Eq. (29). Using the condition (11) for $J_{\pm}$, one obtains

$$J_{\pm}^{\dagger}(x, \lambda^*) = J_{\pm}^{-1}(x, \lambda). \tag{57}$$

From (24) and (32), together with (20), the following relationship can be obtained

$$\left(P^{+}\right)^{\dagger}\left(\lambda^*\right) = P^{-}(\lambda), \tag{58}$$

and we can also find that

$$S^{\dagger}(\lambda^*) = R(\lambda) = S^{-1}(\lambda). \tag{59}$$

Now we turn to the t-evolution of the scattering data. Consider $J_{\pm}$ satisfying Eq. (8). Multiplying both sides of (8) by E, we see that $J_{+}ES$ also satisfies (8). Combined with the boundary condition (11) and the fact that $Q_0 \to 0$ as $x \to \pm\infty$, one can shows that

$$S_t = (4\beta\lambda^3\mathrm{i} - \alpha\lambda^2\mathrm{i} + \frac{\mathrm{i}}{\lambda+\omega})[\sigma_3, S]. \tag{60}$$

Similarly,

$$R_t = (4\beta\lambda^3\mathrm{i} - \alpha\lambda^2\mathrm{i} + \frac{\mathrm{i}}{\lambda+\omega})[\sigma_3, R]. \tag{61}$$

Hence, the following equations hold

$$\frac{\partial s_{22}}{\partial t} = \frac{\partial s_{11}}{\partial t} = 0, \tag{62}$$

and

$$\frac{\partial s_{12}}{\partial t} = (8\beta\lambda^3\mathrm{i} - 2\alpha\lambda^2\mathrm{i} + \frac{2\mathrm{i}}{\lambda+\omega})s_{12}, \tag{63}$$

$$\frac{\partial s_{21}}{\partial t} = -(8\beta\lambda^3\mathrm{i} - 2\alpha\lambda^2\mathrm{i} + \frac{2\mathrm{i}}{\lambda+\omega})s_{21}. \tag{64}$$

Consequently, one gets the following expression

$$s_{12}(t;\lambda) = s_{12}(0;\lambda)e^{(8\beta\lambda^3\mathrm{i}-2\alpha\lambda^2\mathrm{i}+\frac{2\mathrm{i}}{\lambda+\omega})t}, \tag{65}$$

$$s_{21}(t;\lambda) = s_{21}(0;\lambda)e^{-(8\beta\lambda^3\mathrm{i}-2\alpha\lambda^2\mathrm{i}+\frac{2\mathrm{i}}{\lambda+\omega})t}. \tag{66}$$

At this point, we have completed the analysis of the scattering data, and we learn that the RHP can be regarded as being in $\forall(x,t) \in \mathbb{R}\times\mathbb{R}$.

2.3 Solution of the RHP

For Lemma 1, the case where $\det P(\lambda) \not\equiv 0$ is called regular, and the opposite case non-regular. We proceed to solve it according to the different cases.

2.3.1 Solution to the regular RHP

If $\det P^{\pm}(\lambda) \neq 0$ in the upper and lower complex half-planes, then the solution to the regular Lemma 1 is unique. Indeed, suppose that the Lemma 1 admits another solution $\tilde{P}^{\pm}$. Then

$$P^{+}(\lambda)(\tilde{P}^{+})^{-1}(\lambda) = (\tilde{P}^{-})^{-1}(\lambda)P^{-}(\lambda), \quad \lambda \in \mathbb{R}. \tag{67}$$

Since $\det P^{\pm}(\lambda) \neq 0$ and $\det \tilde{P}^{\pm}(\lambda) \neq 0$ for $\lambda \in \mathbb{C}_{\pm}$, respectively, it follows that $P^{+}(\tilde{P}^{+})^{-1}$ is analytic in $\mathbb{C}_{+}$ and $(P^{-})^{-1}(\tilde{P}^{-})$ is analytic in $\mathbb{C}_{-}$. Combining this with (67), exist a matrix-valued function. It is analytic on the entire complex plane. In view of the normalization condition (43) and Liouville's theorem, we obtain

$$P^{+}(\lambda)(\tilde{P}^{+})^{-1}(\lambda) = (\tilde{P}^{-})^{-1}(\lambda)P^{-}(\lambda) = I, \quad \lambda \in \mathbb{C}. \tag{68}$$

Consequently,

$$P^{\pm}(\lambda) = \tilde{P}^{\pm}(\lambda), \tag{69}$$

which proves the uniqueness of the solution to the regular Lemma 1.

Next, we rewrite the jump condition (41) as

$$\left(P^{+}\right)^{-1}(\lambda) - P^{-}(\lambda) = \widehat{G}(\lambda)\left(P^{+}\right)^{-1}(\lambda), \quad \lambda \in \mathbb{R}, \tag{70}$$

where

$$\widehat{G}(\lambda) = -E\begin{pmatrix} 0 & r_{12} \\ s_{21} & 0 \end{pmatrix}E^{-1}. \tag{71}$$

Then, by applying Plemelj's formula together with the normalization condition (43), the unique solution to the Lemma 1 is given by the following integral equation:

$$(P^{+})^{-1}(\lambda) = I + \frac{1}{2\pi\mathrm{i}}\int_{-\infty}^{\infty}\frac{\widehat{G}(\xi)(P^{+})^{-1}(\xi)}{\xi-\lambda}\,d\xi, \quad \lambda \in \mathbb{C}_{+}. \tag{72}$$

2.3.2 Solution of the non-regular RHP

Turn to the case where the Lemma 1 is non-regular. Suppose that there exist points $\lambda_k \in \mathbb{C}_{+}$ and $\hat{\lambda}_k \in \mathbb{C}_{-}$ $(1 \le k \le N)$ such that

$$\det P^{+}(\lambda_k) = 0, \quad \det P^{-}(\hat{\lambda}_k) = 0, \tag{73}$$

the number of zeros is denoted by N, moreover, the symmetry condition (59) implies that

$$\lambda_k^* = \hat{\lambda}_k. \tag{74}$$

The kernels of $P^{\pm}(\lambda_k)$ are spanned by the column vector v_k and the row vector $\hat{v}_k$, respectively; i.e.,

$$P^+(\lambda_k)v_k = 0, \quad \hat{v_k}P^-(\hat{\lambda_k}) = 0, \quad 1 \le k \le N. \tag{75}$$

Also, from (58) to (75), we obtain

$$\hat{v}_k = v_k^{\dagger}. \tag{76}$$

We recall the results for non-regular RHPs due to Zakharov and Shabat [44], which are stated in the following Lemma.

Lemma 2 *Under the standard regularity and normalization conditions, the solution to the non-regular RHP with zeros is given by:*

$$P^+(\lambda) = \widehat{P}^+(\lambda)\Gamma(\lambda), \tag{77}$$
$$P^-(\lambda) = \Gamma^{-1}(\lambda)\widehat{P}^-(\lambda), \tag{78}$$

where

$$\Gamma(\lambda) = I + \sum_{j,k=1}^{N} \frac{|v_j\rangle (M^{-1})_{jk} \langle \hat{v_k}|}{\lambda - \hat{\lambda_k}}, \tag{79}$$

$$\Gamma^{-1}(\lambda) = I - \sum_{j,k=1}^{N} \frac{|v_j\rangle (M^{-1})_{jk} \langle \hat{v_k}|}{\lambda - \lambda_j}. \tag{80}$$

Consider a matrix $M \in \mathbb{C}^{N\times N}$ whose entries are:

$$M_{jk} = \frac{\langle \hat{v_j}|v_k\rangle}{\hat{\lambda_j} - \lambda_k}, \quad 1 \le j,k \le N, \tag{81}$$

$$\det \Gamma(\lambda) = \prod_{k=1}^{N} \frac{\lambda - \lambda_k}{\lambda - \hat{\lambda_k}}. \tag{82}$$

From the preceding theorem, $\widehat{P}^{\pm}(\lambda)$ is the only solution to the regular RHP:

$$\hat{P}^-(\lambda)\hat{P}^+(\lambda) = \Gamma(\lambda)G(\lambda)\Gamma^{-1}(\lambda), \quad \lambda \in \mathbb{R}, \tag{83}$$

where $\hat{P}^{\pm}(\lambda)$ is analytic in $\mathbb{C}_{\pm}$, and satisfies $\hat{P}^{\pm}(\lambda) \to I$, as $\lambda \to \infty$.

Differentiating (75) with respect to x, and noting that P^+ satisfies (7), we obtain

$$P^+(\lambda_k; x)\left(\frac{dv_k}{dx} + \mathrm{i}\lambda_k\sigma_3 v_k\right) = 0. \tag{84}$$

Combining this with (75) yields

$$\frac{dv_k}{dx} + \mathrm{i}\lambda_k\sigma_3 v_k = \alpha_k(x)v_k. \tag{85}$$

Solving the above equation gives

$$v_k(x) = e^{-\mathrm{i}\lambda_k x\sigma_3} v_{k0} e^{\int_{x_0}^{x} \alpha_k(y)dy}, \tag{86}$$

where $v_{k0} = v_k|_{x=0}$. Substituting the obtained solution into Eqs. (79)—(82), we find that the term $e^{\int_{x_0}^{x} \alpha_k(y)dy}$ can be omitted without affecting the result. Therefore, the expression for the vector v_k reduces to

$$v_k(x) = e^{-\mathrm{i}\lambda_k x\sigma_3} v_{k0}. \tag{87}$$

Similarly, by repeating the above procedure, we obtain

$$v_k(t) = e^{\mathrm{i}(4\beta\lambda_k^3 - \alpha\lambda_k^2 + \frac{1}{\lambda_k+\omega})t\sigma_3} v_{k1}. \tag{88}$$

Therefore, the space-time dependence of v_k is as follows:

$$v_k(x,t) = e^{-\mathrm{i}\lambda_k x\sigma_3 + \mathrm{i}(4\beta\lambda_k^3 - \alpha\lambda_k^2 + \frac{1}{\lambda_k+\omega})t\sigma_3} v_{k,0}. \tag{89}$$

Similarly, the expression for $\hat{v}_k$ can be obtained as follows:

$$\hat{v}_k(x,t) = v_{k,0}^{\hat{}} e^{\mathrm{i}\hat{\lambda}_k x\sigma_3 - \mathrm{i}(4\beta\hat{\lambda}_k^3 - \alpha\hat{\lambda}_k^2 + \frac{1}{\hat{\lambda}_k+\omega})t\sigma_3}. \tag{90}$$

3 Asymptotic behavior of the solution in the long-time regime

The Eq. (1) have no explicit solutions at any later time t. However, their long-time asymptotic behavior are characterized. According to the solution of conventional Lemma 1, the solution of Eq. (83) admits the following integral representation:

$$\left(\widehat{P}^+\right)^{-1}(\lambda) = I + \frac{1}{2\pi\mathrm{i}} \int_{-\infty}^{\infty} \frac{\Gamma(\xi)\widehat{G}(\xi)\Gamma^{-1}(\xi)\left(\widehat{P}^+\right)^{-1}(\xi)}{\xi - \lambda} d\xi, \lambda \in \mathbb{C}_+, \tag{91}$$

where $\hat{G} = I - G$. As $\lambda \to \infty$

$$(\hat{P}^+)^{-1}(\lambda) \to I - \frac{1}{2\pi\mathrm{i}\lambda} \int_{-\infty}^{\infty} \Gamma(\xi)\hat{G}(\xi)\Gamma^{-1}(\xi)(\hat{P}^+)^{-1}(\xi)d\xi, \tag{92}$$

and thus

$$\hat{P}^+(\lambda) \to I + \frac{1}{2\pi\mathrm{i}\lambda}$$

$$\int_{-\infty}^{\infty} \Gamma(\xi)\hat{G}(\xi)\Gamma^{-1}(\xi)(\hat{P}^{+})^{-1}(\xi)d\xi. \tag{93}$$

According to Eq. (79)

$$\Gamma(\lambda) \to I + \frac{1}{\lambda}\sum_{j,k=1}^{N} |v_j\rangle (M^{-1})_{jk}\langle \hat{v_k}|, \quad \lambda \to \infty. \tag{94}$$

From Eqs. (77), (93), and (94), it follows that the term P_1^+ in the expansion of P^+ is given by

$$P_1^+(x,t) = \sum_{j,k=1}^{N} |v_j\rangle (M^{-1})_{jk}\langle \hat{v}_k| + \frac{1}{2\pi \mathrm{i}} \int_{-\infty}^{\infty} \Gamma(\xi)\widehat{G}(\xi)\Gamma^{-1}(\xi)(\widehat{P}^{+})^{-1}(\xi)d\xi. \tag{95}$$

Using Eq. (71), together with Eqs. (12) and (65), we derive

$$\widehat{G}(\xi;x,t) = \begin{pmatrix} 0 & -s_{12}(0;\xi) e^{-2\mathrm{i}\xi x + \mathrm{i}(8\beta\xi^3 - 2\alpha\xi^2 + \frac{2}{\xi+\omega})t} \\ -s_{21}(0;\xi) e^{2\mathrm{i}\xi x - \mathrm{i}(8\beta\xi^3 - 2\alpha\xi^2 + \frac{2}{\xi+\omega})t} & 0 \end{pmatrix}. \tag{96}$$

Setting $\hat{\xi} = \xi\sqrt[3]{t}$ and substituting it into Eq. (95) yields

$$\int_{-\infty}^{\infty} \Gamma(\xi)\widehat{G}(\xi)\Gamma^{-1}(\xi)\left(\widehat{P}^{+}\right)^{-1}(\xi)d\xi \to 0, \quad t \to \infty. \tag{97}$$

Furthermore, we know

$$P_1^+(x,t) \to \sum_{j,k=1}^{N} |v_j\rangle (M^{-1})_{jk}\langle \hat{v}_k|, \quad t \to \infty. \tag{98}$$

Finally, the corresponding asymptotic solution is given by

$$u(x,t) \to 2\mathrm{i}\left(\sum_{j,k=1}^{N} |v_j\rangle (M^{-1})_{jk}\langle \hat{v_k}|\right)_{12}, \quad t \to \infty. \tag{99}$$

4 *N*-soliton solutions for the H-MB system

4.1 *N*-soliton solutions expression

When $G = I$ and $\hat{G} = 0$, Eq. (95) admits the solution

$$P_1^+(x,t) = \sum_{j,k=1}^{N} |v_j\rangle (M^{-1})_{jk}\langle \hat{v}_k|. \tag{100}$$

Introduce

$$\theta_k(x,t,\lambda_k) = -\mathrm{i}\lambda_k x + \mathrm{i}\left(4\beta\lambda_k^3 - \alpha\lambda_k^2 + \frac{1}{\lambda_k+\omega}\right)t, \tag{101}$$

then (89) and (90) can be rewritten as

$$v_k = e^{\theta_k\sigma_3}v_{k,0}, \tag{102}$$

$$\hat{v}_k = \hat{v}_{k,0}e^{\hat{\theta}_k\sigma_3}, \tag{103}$$

where $v_{k,0} = (\alpha_k, \beta_k)^T$ for $k = 1, 2, \ldots, N$, with $v_{k,0}$ denoting a fixed vector in the complex domain. Consequently, we can obtain

$$u(x,t) = -2\mathrm{i}\sum_{j,k=1}^{N} \alpha_j\beta_k^* e^{\theta_j - \theta_k^*}(M^{-1})_{jk}, \tag{104}$$

where M is an N-order matrix whose entries are

$$M_{jk} = \frac{1}{\lambda_k - \hat{\lambda}_j}\left[\alpha_k\alpha_j^* e^{\theta_j^* + \theta_k} + \beta_k\beta_j^* e^{-(\theta_j^* + \theta_k)}\right]. \tag{105}$$

This yields as following:

$$u(x,t) = 2\mathrm{i}\frac{\det F}{\det M}, \tag{106}$$

where the matrix F is an $(N+1)$-dimensional defined by

$$F = \begin{pmatrix} 0 & \beta_1^* e^{-\theta_1^*} & \ldots & \beta_N^* e^{-\theta_N^*} \\ \alpha_1 e^{\theta_1} & M_{11} & \ldots & M_{N1} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_N e^{\theta_N} & M_{1N} & \ldots & M_{NN} \end{pmatrix}. \tag{107}$$

4.2 Single-soliton solutions

For the case $N = 1$, following [45], we set $v_{1,0} = (\alpha_1, \beta_1)^T$ with $\alpha_1^*\alpha_1 > 0$ and $\beta_1^*\beta_1 > 0$. Define

$$\lambda_1 = a + \mathrm{i}b, \quad \frac{\beta_1^*\beta_1}{\alpha_1^*\alpha_1} = e^{-2\eta_1}, \tag{108}$$

then the solution u is given by

$$u(x,t) = 2\mathrm{i}(\hat{\lambda}_1 - \lambda_1)\frac{\alpha_1\beta_1^* e^{\theta_1 - \theta_1^*}}{|\alpha_1|^2 e^{\theta_1 + \theta_1^*} + |\beta_1|^2 e^{-(\theta_1 + \theta_1^*)}}, \tag{109}$$

which can be rewritten as

$$u = 2b\frac{\beta_1^*}{\alpha_1^*}e^{\eta_1 + \theta_1 - \theta_1^*}\,\mathrm{sech}\left(\theta_1^* + \theta_1 + \eta_1\right). \tag{110}$$

Also, from the definition of θ_k, we obtain

$$\theta_1+\theta_1^*=2b\left[x-\left(12\beta a^2-4\beta b^2-2\alpha a-\frac{1}{(a+\omega)^2+b^2}\right)t\right], \tag{111}$$

$$\theta_1-\theta_1^*=2\mathrm{i}\left[-ax+\left(4\beta a\left(a^2-3b^2\right)-\alpha\left(a^2-b^2\right)+\frac{a+\omega}{(a+\omega)^2+b^2}\right)t\right]. \tag{112}$$

Thus, Eq. (110) can be reformulated as follows

$$u(x,t)=2b\,\mathrm{sech}\,[m(x,t)]\,e^{\mathrm{i}\chi(x,t)}, \tag{113}$$

$$p(x,t)=\frac{2\mathrm{i}b(a+\omega)\cosh(m(x,t))-2b^2\sinh(m(x,t))}{\left[(a+\omega)^2+b^2\right]\cosh^2(m(x,t))}e^{\mathrm{i}\chi(x,t)}, \tag{114}$$

$$\eta(x,t)=\frac{\left[(a+\omega)^2+b^2\right]\cosh^2[m(x,t)]-2b^2}{\left[(a+\omega)^2+b^2\right]\cosh^2[m(x,t)]}, \tag{115}$$

where

$$m(x,t)=2b\left[x-\left(12\beta a^2-4\beta b^2-2\alpha a-\frac{1}{(a+\omega)^2+b^2}\right)t\right]+\eta_1,$$

$$\chi(x,t)=2\left[-ax+\left(4\beta a\left(a^2-3b^2\right)-\alpha\left(a^2-b^2\right)+\frac{a+\omega}{(a+\omega)^2+b^2}\right)t\right].$$

From the explicit Eqs. (113)–(115), the fundamental physical properties of the single-soliton solution, such as amplitude, velocity, and center position can be explicitly identified. According to Eq. (113), the amplitude of the envelope $|u|$ determined by the imaginary part of the spectral parameter λ_1 is $2b$. From the expression of $m(x,t)$, we obtain the velocity of the soliton is not only dependent on the spectral parameters a and b, but also nonlinearly modulated by the system parameters α, β and the background frequency ω. Thus the expression is

$$z=12\beta a^2-4\beta b^2-2\alpha a-\frac{1}{(a+\omega)^2+b^2}.$$

Furthermore, the center position of the soliton pulse at any time t can be expressed as:

$$x_c(t)=zt-\frac{\eta_1}{2b},$$

where $-\frac{\eta_1}{2b}$ represents the initial position of the soliton peak at $t=0$. These characteristic parameters provide a theoretical basis for analyzing the asymptotic behavior and phase shifts during multi-soliton interactions.

Figures 1 and 2 present a comprehensive characterization of the single-soliton solution for the H-MB system under specific parameter values, using three-dimensional plots. The three-dimensional of $|u|$, $|p|$ and η revealing a stable, localized wave packet whose shape and velocity remain unchanged, a defining characteristic of solitons.

In Fig. 1, the population inversion component η exhibits a typical dark soliton profile as shown. Analytically, Eq. (115) reveals that η approaches the constant background value 1 as $|x|\to\infty$. At the soliton center $m(x,t)=0$, the value of η drops to its minimum

$$\eta_{\min}=\frac{(a+\omega)^2-b^2}{(a+\omega)^2+b^2},$$

this localized dip on a non-zero background characterizes a dark soliton.

In Fig. 2 the double-peak intensity profile of the polarization component $|p|$ can be analytically explained by analyzing the modulus of Eq. (114). One can derive that

$$|p|\propto\sqrt{(A^2+b^2)\mathrm{sech}^2(m)-b^2\mathrm{sech}^4(m)},$$

where $A=a+\omega$. Further extremum analysis reveals that when the criterion $b^2>(a+\omega)^2$ is satisfied, the maximum of $|p|$ shifts away from the center $m(x,t)=0$, resulting in a symmetric double-peak structure within a single soliton. This intrinsic morphological property of the single-soliton solution provides a fundamental explanation for the double-peak distribution observed in the subsequent two-soliton collision scenarios.

4.3 Two-soliton solutions

For the case $N=2$, with $v_{k,0}=(\alpha_k,\beta_k)^T$ for $k=1,2$, the two-soliton solution of system (1) can be obtained explicitly. Define

$$\lambda_1=a_1+\mathrm{i}b_1,\quad \lambda_2=a_2+\mathrm{i}b_2, \tag{116}$$

the two-soliton solution for u is then given by

$$u(x,t)=-\frac{2\mathrm{i}}{A}\left(\alpha_1\beta_1^*e^{\theta_1-\theta_1^*}M_{22}-\alpha_1\beta_2^*e^{\theta_1-\theta_2^*}M_{12}-\alpha_2\beta_1^*e^{\theta_2-\theta_1^*}M_{21}+\alpha_2\beta_2^*e^{\theta_2-\theta_2^*}M_{11}\right), \tag{117}$$

where

$$A=M_{11}M_{22}-M_{12}M_{21},\quad A\neq 0,$$

and the matrix entries are

$$M_{11}=\frac{\alpha_1\alpha_1^*e^{\theta_1+\theta_1^*}+\beta_1\beta_1^*e^{-\theta_1-\theta_1^*}}{\lambda_1-\hat{\lambda}_1},$$

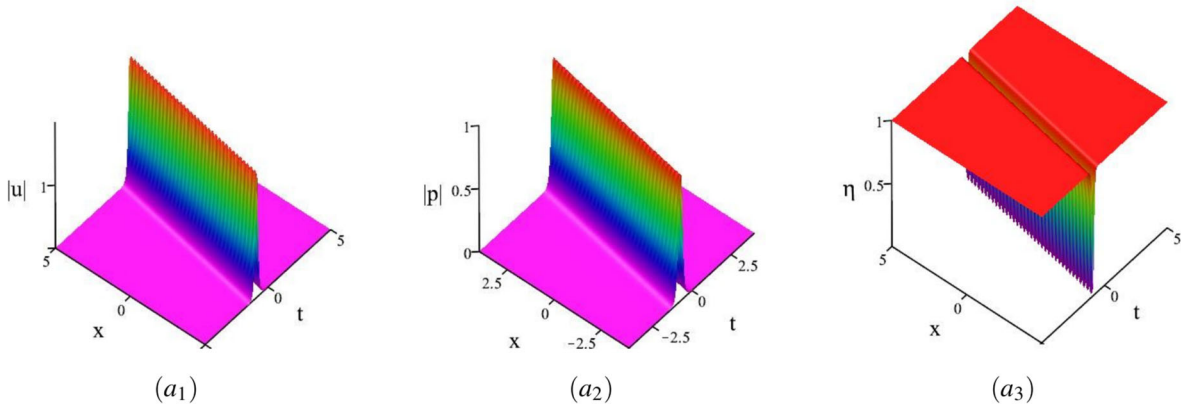


Fig. 1 The three-dimensional plots of one soliton solution $|u|$, $|p|$ and η with $\alpha = 1, \beta = 1, \omega = 0, \alpha_1 = 1, \beta_1 = 1, \eta_1 = 0, a = 1, b = 1$. Among them, $|u|$ is bright solitons, $|p|$ is bright soliton, and η is a dark soliton

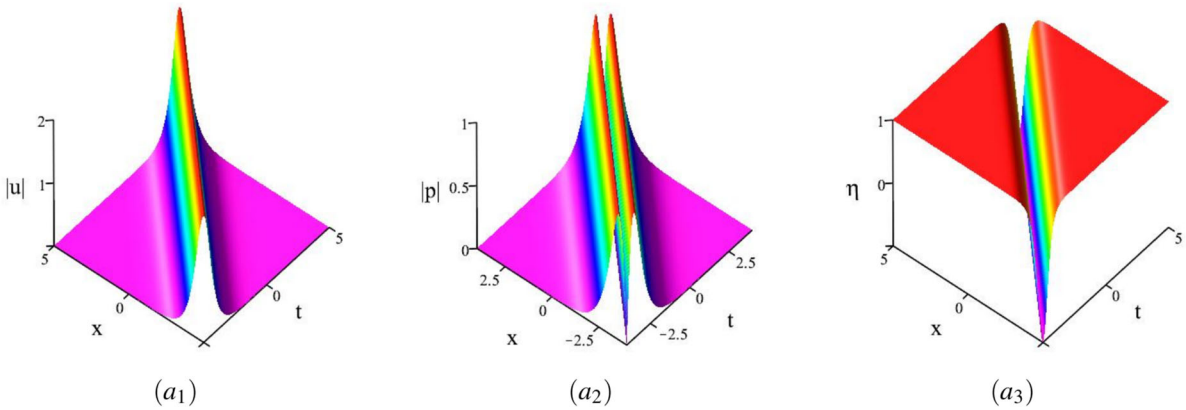


Fig. 2 The three-dimensional plots of $|u|$, $|p|$ and η when $N = 1$ with $\alpha = -1, \beta = -\frac{1}{2}, \omega = 0, \alpha_1 = 1, \beta_1 = 1, \eta_1 = 0, a = 0, b = 1$. Among them, $|u|$ is bright solitons, $|p|$ is bright soliton with double-peak, and η is a dark soliton

$$M_{12} = \frac{\alpha_2\alpha_1^* e^{\theta_2+\theta_1^*} + \beta_2\beta_1^* e^{-\theta_2-\theta_1^*}}{\lambda_2 - \hat{\lambda}_1},$$
$$M_{21} = \frac{\alpha_1\alpha_2^* e^{\theta_1+\theta_2^*} + \beta_1\beta_2^* e^{-\theta_1-\theta_2^*}}{\lambda_1 - \hat{\lambda}_2},$$
$$M_{22} = \frac{\alpha_2\alpha_2^* e^{\theta_2+\theta_2^*} + \beta_2\beta_2^* e^{-\theta_2-\theta_2^*}}{\lambda_2 - \hat{\lambda}_2}.$$

Figures 3, 4 and 5 systematically present two typical types of behavior of the H-MB system in the two-soliton scenario: colliding states and bound states. Figure 3 correspond to the elastic collision process. Specifically, Fig. 3 shows the three-dimensional evolution of $|u|$ (bright soliton), $|p|$ (bright soliton with double peaks) and η (dark soliton) during the collision, intuitively illustrating the elastic collision feature in which two solitons approach each other from opposite sides, pass through one another, and maintain their shapes and velocities. Figures 4 and 5 correspond to the bound state case, where the two solitons share the same velocity and remain bound together during propagation. Therefore, the bound state of two solitons should satisfy the following conditions

$$(a_1 - a_2)\left[12\beta(a_1 + a_2) - 2\alpha + \frac{a_1 + a_2 + 2\omega}{B}\right]$$
$$-(b_1 - b_2)\left[4\beta(b_1 + b_2) - \frac{b_1 + b_2}{B}\right] = 0,$$

where $B = [(a_1 + \omega)^2 + b_1^2][(a_2 + \omega)^2 + b_2^2]$.

In the first case, i.e., when $z_1 \neq z_2$, the dynamical properties of the solitons are shown in Fig. 3. As time t tends to $-\infty$, the two solitons are far apart and then slowly approach each other. Upon collision, a strong interaction occurs. As t tends to $+\infty$, the two solitons emerge from the collision again, retaining the same shape and speed, and without any energy loss as they move apart. Therefore, the interaction between these solitons is elastic.

The dynamic behavior is categorized into two cases depending on the velocity hierarchy. In the regime $z_1 < z_2$, the slower soliton-1 is initially positioned to the right of soliton-2 as $t \to -\infty$. In this case, note that $b_k > 0$ for $\lambda_k \in \mathbb{C}_+$. Alternatively, for the case $z_1 > z_2$, at $t = -\infty$,

soliton-1 lies to the left of soliton-2 and travels at a higher speed, with $b_k < 0$ holding for $\lambda_k \in \mathbb{C}_-$.

Within the moving frame traveling at velocity z_1, we define

$$\begin{aligned} \mathrm{Re}(\theta_1) &= b_1 x - \left[4\beta(3a_1^2 b_1 - b_1^3) - 2\alpha a_1 b_1 - \frac{b_1}{(a_1+\omega)^2 + b_1^2} \right] \\ t &= b_1(x - z_1 t) = O(1), \end{aligned} \tag{118}$$

where

$$z_1 = 4\beta(3a_1^2 - b_1^2) - 2\alpha a_1 - \frac{1}{(a_1+\omega)^2 + b_1^2}. \tag{119}$$

Since

$$\mathrm{Re}(\theta_2) = b_2(x - z_2 t) = b_2(x - z_1 t) + b_2(z_1 - z_2)t, \tag{120}$$

with

$$z_2 = 4\beta(3a_2^2 - b_2^2) - 2\alpha a_2 - \frac{1}{(a_2+\omega)^2 + b_2^2}, \tag{121}$$

when $t \to -\infty$, $\mathrm{Re}(\theta_2) \to \infty$, the asymptotic expression of the two-soliton becomes

$$u(x,t) \to 2\mathrm{i}(\hat{\lambda}_1 - \lambda_1) \frac{\alpha_1^- \beta_1^* e^{\theta_1 - \theta_1^*}}{\left|\alpha_1^-\right|^2 e^{\theta_1 + \theta_1^*} + |\beta_1|^2 e^{-(\theta_1 + \theta_1^*)}}, \tag{122}$$

where $\alpha_1^- = \alpha_1(\lambda_1 - \lambda_2)/(\lambda_1 - \hat{\lambda}_2)$. This asymptotic solution corresponds to a single soliton characterized by peak amplitude $2b\frac{\beta_1^*}{\alpha_1^*}$ and velocity z_1.

As $t \to \infty$ and $\mathrm{Re}(\theta_2) \to -\infty$, the two-soliton solution reduces to the asymptotic form

$$u(x,t) \to 2\mathrm{i}(\hat{\lambda}_1 - \lambda_1) \frac{\alpha_1^+ \beta_1^* e^{\theta_1 - \theta_1^*}}{\left|\alpha_1^+\right|^2 e^{\theta_1 + \theta_1^*} + |\beta_1|^2 e^{-(\theta_1 + \theta_1^*)}}, \tag{123}$$

where $\alpha_1^+ = \alpha_1(\lambda_1 - \hat{\lambda}_2)/(\lambda_1 - \lambda_2)$. The asymptotic single-soliton solution in Eq. (123) possesses velocity z_1 and peak amplitude $2b_1\frac{\beta_1^*}{\alpha_1^*}$. This indicates that while the soliton's profile and speed remain invariant after the interaction, it undergoes discrete offsets in both phase and position. The position shift is given by

$$\begin{aligned} \Delta x_{01} &= -\frac{\alpha_1^*}{2b_1\beta_1^*} \left(\ln|\alpha_1^+| - \ln|\alpha_1^-| \right) \\ &= \frac{\alpha_1^*}{b_1\beta_1^*} \ln\left| \frac{\lambda_1 - \lambda_2}{\lambda_1 - \hat{\lambda}_2} \right|, \end{aligned} \tag{124}$$

and the phase shift is

$$\Delta\sigma_{01} = \arg(\alpha_1^+) - \arg(\alpha_1^-) = -2\arg\left(\frac{\lambda_1 - \lambda_2}{\lambda_1 - \hat{\lambda}_2} \right). \tag{125}$$

Within the moving frame traveling at velocity z_2, as $t \to \pm\infty$, the asymptotic solutions each reduce to a single soliton characterized by peak amplitude $2b_2\frac{\beta_2^*}{\alpha_2^*}$ and velocity z_2. Consequently, the interaction induces a position shift for the second soliton:

$$\Delta x_{02} = \frac{\alpha_2^*}{b_2\beta_2^*} \ln\left| \frac{\lambda_2 - \lambda_1}{\lambda_2 - \hat{\lambda}_1} \right|, \tag{126}$$

and a phase shift:

$$\Delta\sigma_{01} = -2\arg\left(\frac{\lambda_2 - \lambda_1}{\lambda_2 - \hat{\lambda}_1} \right). \tag{127}$$

Next, we consider the second case, where $z_1 = z_2$, this is called a bound state. In particular, we consider two special cases of $a_1 = a_2$ and $b_1 = b_2$. The behavior of the solutions in this case are illustrated respectively in Figs. 4 and 5. In this situation, the two constituent solitons share identical velocities and consequently remain together while propagating at a common speed.

4.4 Three-soliton solutions

For N=3, the three-soliton solution can get by (106). Define

$$\lambda_1 = a_1 + \mathrm{i}b_1, \quad \lambda_2 = a_2 + \mathrm{i}b_2, \quad \lambda_3 = a_3 + \mathrm{i}b_3, \tag{128}$$

the solution for u is as follows:

$$u(x,t) = 2\mathrm{i}\frac{F_3}{M_3}, \tag{129}$$

where

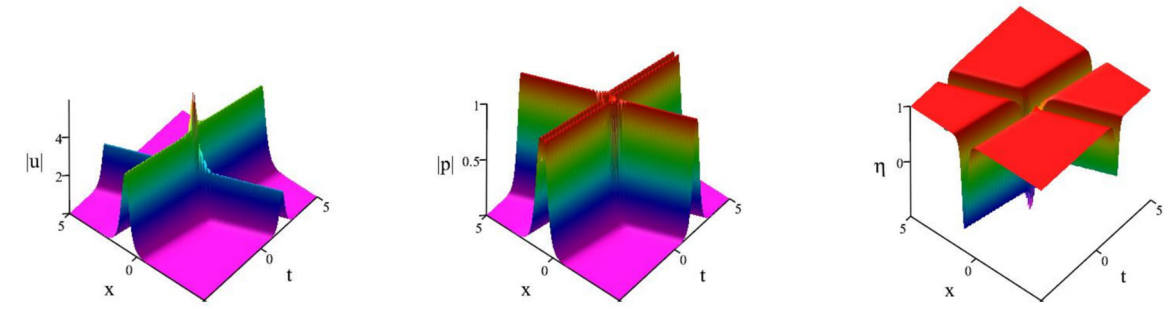
$$\begin{aligned} F_3 = &-\beta_1^* e^{-\theta_1^*}\Big[\alpha_1 e^{\theta_1}(M_{22}M_{33} - M_{23}M_{32}) \\ &+ \alpha_2 e^{\theta_2}(M_{23}M_{31} - M_{21}M_{33}) + \alpha_3 e^{\theta_3}(M_{21}M_{32} - M_{31}M_{22})\Big] \\ &+ \beta_2^* e^{-\theta_2^*}\Big[\alpha_1 e^{\theta_1}(M_{12}M_{33} - M_{13}M_{32}) + \alpha_2 e^{\theta_2}(M_{13}M_{31} - M_{11}M_{33}) \\ &+ \alpha_3 e^{\theta_3}(M_{11}M_{32} - M_{12}M_{31})\Big] \\ &- \beta_3^* e^{-\theta_3^*}\Big[\alpha_1 e^{\theta_1}(M_{12}M_{23} - M_{13}M_{22}) + \alpha_2 e^{\theta_2}(M_{13}M_{21} - M_{11}M_{23}) \\ &+ \alpha_3 e^{\theta_3}(M_{11}M_{22} - M_{12}M_{21})\Big], \end{aligned}$$

and

$$\begin{aligned} M_3 = &M_{11}M_{22}M_{33} + M_{21}M_{32}M_{13} + M_{31}M_{12}M_{23} \\ &- M_{13}M_{22}M_{31} - M_{12}M_{21}M_{33} - M_{11}M_{32}M_{23}, \end{aligned}$$

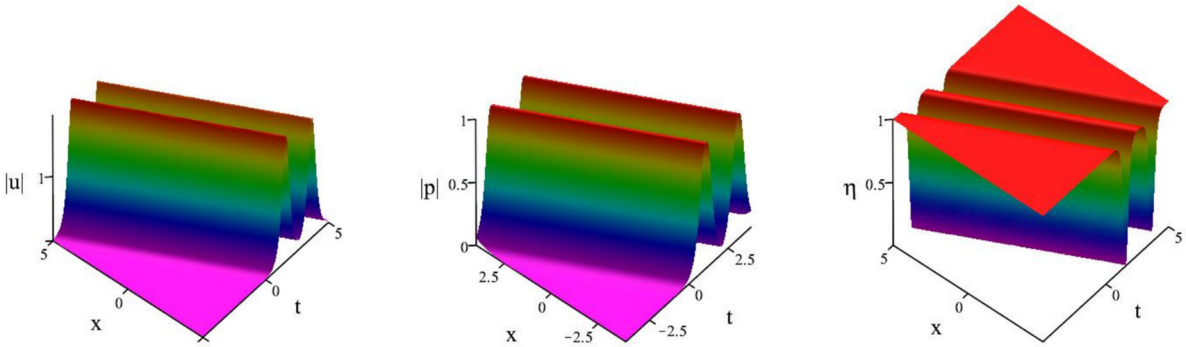
with the matrix entries

$$M_{11} = \frac{\alpha_1\alpha_1^* e^{\theta_1 + \theta_1^*} + \beta_1\beta_1^* e^{-\theta_1 - \theta_1^*}}{\lambda_1 - \hat{\lambda}_1},$$



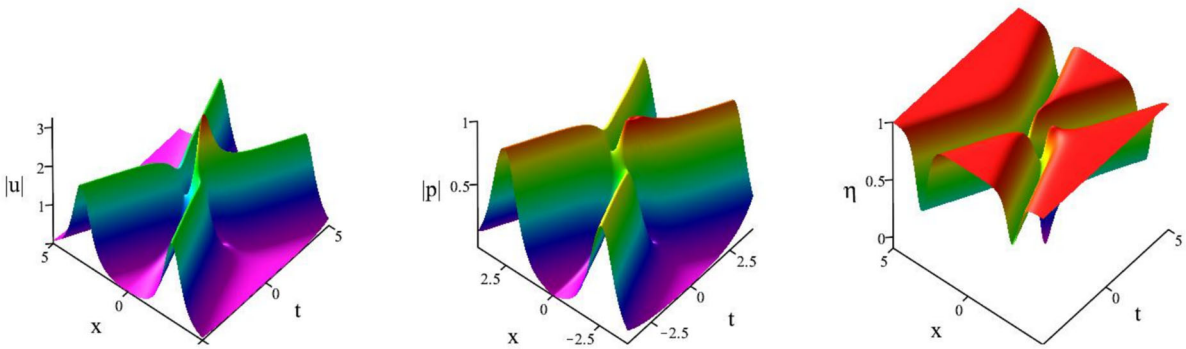
(a_1) (a_2) (a_3)

Fig. 3 The three-dimensional of $|u|$, $|p|$ and η under $N = 2$ with $\alpha = -1$, $\beta = -\frac{1}{2}$, $\omega = 0$, $\alpha_1 = I$, $\beta_1 = 1$, $\alpha_2 = 1$, $\beta_2 = 1$, $a_1 = 1$, $b_1 = 1$, $a_2 = -1$, $b_2 = 2$. Among them, $|u|$ is bright solitons, $|p|$ is bright soliton with double-hump, and η is a dark soliton



(a_1) (a_2) (a_3)

Fig. 4 Pictures are bound state of two-soliton solution when $a_1 = a_2$, the three-dimensional plots of $|u|$, $|p|$ and η with $\alpha = 1$, $\beta = \frac{18}{265}$, $\omega = 0$, $\alpha_1 = 1 + I$, $\beta_1 = 1$, $\alpha_2 = 2 - \frac{I}{2}$, $\beta_2 = 1$, $a_1 = 1$, $b_1 = 1$, $a_2 = 1$, $b_2 = -\frac{11}{12}$. Among them, $|u|$ is bright solitons, $|p|$ is bright soliton, and η is a dark soliton



(a_1) (a_2) (a_3)

Fig. 5 Pictures are bound state of two-soliton solution when $b_1 = b_2$, three-dimensional plots of $|u|$, $|p|$ and η with $\alpha = \frac{7}{65}$, $\beta = \frac{58}{585}$, $\omega = 1$, $\alpha_1 = 1 + I$, $\beta_1 = 1$, $\alpha_2 = 2 - \frac{1}{2}I$, $\beta_2 = 1$, $a_1 = \frac{1}{2}$, $b_1 = 1$, $a_2 = 1$, $b_2 = 1$, Among them, $|u|$ and $|p|$ are bright soliton, η is a dark soliton.

$$M_{12}=\frac{\alpha_2\alpha_1^* e^{\theta_2+\theta_1^*}+\beta_2\beta_1^* e^{-\theta_2-\theta_1^*}}{\lambda_2-\hat{\lambda}_1},$$
$$M_{13}=\frac{\alpha_3\alpha_1^* e^{\theta_3+\theta_1^*}+\beta_3\beta_1^* e^{-\theta_3-\theta_1^*}}{\lambda_3-\hat{\lambda}_1},$$
$$M_{21}=\frac{\alpha_1\alpha_2^* e^{\theta_1+\theta_2^*}+\beta_1\beta_2^* e^{-\theta_1-\theta_2^*}}{\lambda_1-\hat{\lambda}_2},$$
$$M_{22}=\frac{\alpha_2\alpha_2^* e^{\theta_2+\theta_2^*}+\beta_2\beta_2^* e^{-\theta_2-\theta_2^*}}{\lambda_2-\hat{\lambda}_2},$$
$$M_{23}=\frac{\alpha_3\alpha_2^* e^{\theta_3+\theta_2^*}+\beta_3\beta_2^* e^{-\theta_3-\theta_2^*}}{\lambda_3-\hat{\lambda}_2},$$
$$M_{31}=\frac{\alpha_1\alpha_3^* e^{\theta_1+\theta_3^*}+\beta_1\beta_3^* e^{-\theta_1-\theta_3^*}}{\lambda_1-\hat{\lambda}_3},$$
$$M_{32}=\frac{\alpha_2\alpha_3^* e^{\theta_2+\theta_3^*}+\beta_2\beta_3^* e^{-\theta_2-\theta_3^*}}{\lambda_2-\hat{\lambda}_3},$$
$$M_{33}=\frac{\alpha_3\alpha_3^* e^{\theta_3+\theta_3^*}+\beta_3\beta_3^* e^{-\theta_3-\theta_3^*}}{\lambda_3-\hat{\lambda}_3}.$$

Figure 6 presents the three-dimensional evolution plots of three-soliton $|u|$ under three different parameter sets, intuitively showing the complex interaction process among the three solitons, including their mutual approach, collision, and separation. After the collision, each soliton retains its original shape and velocity, reflecting the essence of elastic collisions. The three parameter sets are following:

The parameter values for subfigure (a_1) are

$$a_1=1, b_1=-\frac{3}{2}, \alpha_1=I, \beta_1=1,$$
$$a_2=-1, b_2=1, \alpha_2=1, \beta_2=1,$$
$$a_3=0, b_3=1, \alpha_3=1, \beta_3=1,$$

the parameter values for subfigure (a_2) are

$$a_1=1, b_1=-1, \alpha_1=I, \beta_1=1,$$
$$a_2=-1, b_2=-2, \alpha_2=1, \beta_2=1,$$
$$a_3=0, b_3=1, \alpha_3=1, \beta_3=1,$$

the parameter values for subfigure (a_3) are

$$a_1=1, b_1=-1, \alpha_1=1+I, \beta_1=1,$$
$$a_2=1, b_2=-\frac{2}{3}, \alpha_2=2-\frac{I}{2}, \beta_2=1,$$
$$a_3=1, b_3=\frac{\sqrt{2}}{2}, \alpha_3=2+I, \beta_3=1+I.$$

5 Infinite hierarchy of conservation laws

Let $Y=(y_1,y_2)^T$ and set $\mu=y_1/y_2$, then it can be readily derived that

$$(\ln y_2)_x=\mathrm{i}\lambda-u^*\mu, \tag{130}$$
$$(\ln y_2)_t=v_3\mu+v_4, \tag{131}$$

where

$$v_3=4\beta u^*\lambda^2-\left(\alpha u^*+2\beta \mathrm{i}u_t^*\right)\lambda-2\beta|u|^2u+\frac{\alpha}{2}\mathrm{i}u_t^*-\beta u_{tt}^*-\frac{\mathrm{i}p^*}{\lambda+\omega},$$
$$v_4=\mathrm{i}\left[-4\beta\lambda^3+\alpha\lambda^2+2\beta|u|^2\lambda-\frac{\alpha}{2}|u|^2-\frac{\eta}{\lambda+\omega}\right]+\beta\left(uu_t^*-u_tu^*\right).$$

From the mixed partials of (130)–(131) yields

$$\left(\mathrm{i}\lambda-u^*\mu\right)_t=(v_3\mu+v_4\mu)_x, \tag{132}$$

i.e.

$$-u_t^*\mu-u^*\mu_t=v_{3,x}\mu+v_3\mu_x+v_{4,x}, \tag{133}$$

where

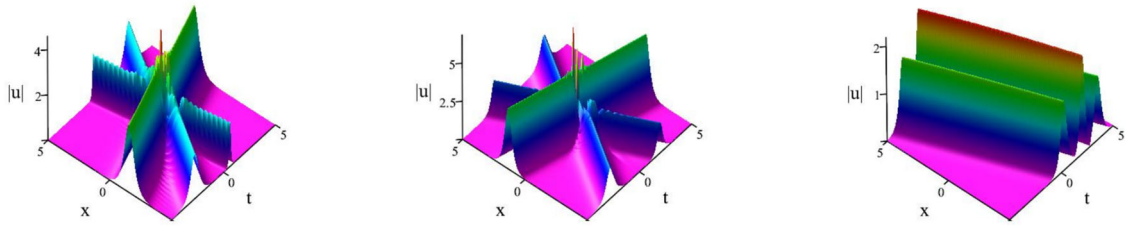
$$v_{3,x}=4\beta u_x^*\lambda^2-\alpha u_x^*\lambda-2\beta \mathrm{i}u_{tx}^*\lambda-2\beta\left(2|u|^2u_x+u_x^*u^2\right)+\frac{\alpha}{2}\mathrm{i}u_{tx}^*-\beta u_{ttx}^*-\frac{\mathrm{i}}{\lambda+\omega}p_x^*,$$
$$v_3=4\beta u^*\lambda^2-\left(\alpha u^*+2\beta \mathrm{i}u_t^*\right)\lambda-2\beta|u|^2u+\frac{\alpha}{2}\mathrm{i}u_t^*-\beta u_{tt}^*-\frac{\mathrm{i}p^*}{\lambda+\omega},$$
$$v_{4,x}=\mathrm{i}\left[\left(2\beta\lambda-\frac{\alpha}{2}\right)\left(u_xu^*+uu_x^*\right)-\frac{\eta_x}{\lambda+\omega}\right]+\beta\left(u_xu_t^*+uu_{tx}^*-u_{tx}u^*-u_tu_x^*\right).$$

Expand μ as a power series in λ^{-1}

$$\mu(x,t,\lambda)=\sum_{n=1}^{\infty}\frac{\mu_n(x,t)}{(2\mathrm{i}\lambda)^n}. \tag{134}$$

Substitute it into Eq. (132), and then compare the same power coefficient of λ^{-1}, we can get

$$(u^*\mu_n)_t=(\beta u^*\mu_{n+2}+\left(\frac{\alpha u^*}{2\mathrm{i}}+\beta u_t^*\right)\mu_{n+1}+\beta u_{tt}^*\mu_n+2\beta|u|^2u\mu_n-\frac{\alpha\mathrm{i}}{2}u_t^*\mu_n-2\eta)_x, \tag{135}$$
$$(u^*\mu_{n+1})_t=(\beta u^*\mu_{n+3}+\left(\frac{\alpha u^*}{2\mathrm{i}}+\beta u_t^*\right)\mu_{n+2}+\beta u_{tt}^*\mu_{n+1}$$



(a_1) (a_2) (a_3)

Fig. 6 The three-dimensional plots of three-soliton solution of $|u|$ under $N = 3$ with different parameter

$$+2\beta|u|^2u\mu_{n+1}-\frac{\alpha \mathrm{i}}{2}u_t^*\mu_{n+1}-2p^*\mu_n)_x, \tag{136}$$

and similarly, we see that μ satisfies the following Riccati equation

$$\mu_x = u^*\mu^2 - 2i\lambda\mu + u, \tag{137}$$

then by substituting (134) into (137), The coefficient of μ_n is given by

$$\mu_1 = u, \quad \mu_2 = -u_x, \tag{138}$$

$$\mu_{n+1} = -\mu_{n,x} + u^*\sum_{k=1}^{n-1}\mu_k\mu_{n-k}, \quad n \geq 2. \tag{139}$$

Based on the above, we substitute Eqs. (138)–(139) into Eqs. (135)–(136), and we can obtain the Eq. (1).

6 Conclusion

This paper presents a systematic and comprehensive investigation of the H-MB system and an account is given of the propagation of femtosecond optical pulses in an erbium-doped fiber system. The IST method, formulated within the RHP framework is employed. This mathematical tool has proven highly effective in the analysis of integrable nonlinear evolution equations, and the present work successfully extends its application to the coupled H-MB system. The investigation commences with the 2×2 matrix spectral problem, i.e., the Lax pair, associated with the H-MB system. Using this approach, we do a spectral analysis and define the Jost solutions as well as the scattering matrix. It is very important to study these things and their symmetry properties carefully for us to develop RHP in the future. In addition, an adjoint spectral problem is introduced to construct the corresponding adjoint Jost solutions, thereby completing the requisite analytical framework. Following this, a matrix RHP is formulated for the H-MB system. A direct and explicit link is established between the solution of this RHP and the potentials appearing in the H-MB system. This formulation offers an alternative to the conventional IST method. In the reflectionless case, where the jump matrix in the RHP reduces to the identity matrix, the explicit one-, two-, and three-soliton solutions of the H-MB system can be obtained (see Figs. 1, 2, 3, 4, 5 and 6). By the constructed spectral problem and the properties of the scattering data, the infinite conservation laws for the H-MB system can also be obtained. This outcome serves as rigorous evidence substantiating the complete integrability of the system.

Furthermore, while some previous researches were restricted to specific parameter choices, our results are valid for general parameters α and β. The parameters α and β are only involved in the phase and velocity terms, implying that they affect the soliton speed while the amplitude remains unchanged. These advancements establish a more robust and versatile theoretical foundation for predicting and controlling nonlinear wave dynamics in complex optical media. The results presented in this work carry several significant implications. From the perspective of mathematical physics, the successful implementation of the RH method is demonstrated for a physically meaningful coupled system, thereby reinforcing its role as a fundamental tool within the theory of integrable systems. From a physical viewpoint, the explicit soliton solutions furnish a theoretical basis for interpreting and potentially controlling the behavior of ultrashort pulses propagating in active optical media.

Author Contributions S. Liu: Writing-original draft, Data curation, Software, Investigation. Y.Q. Liu: Supervision, Validation, Writing-review. All authors reviewed the manuscript.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Conflict of interest The authors declare no conflict of interest.

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